

1. Which statement is true about dynamic websites?

☐ (A) A and C

☐ (B) The content can't be changed anything in the site as it is predefined.

☐ (C) The content is regenerated every time a user visits or reloads

☐ (D) The content of script can be changed at the run time

2. If every non-key attribute is functionally dependent on primary key, then the relation will be in which normal form?

- ☐ **(A)** Fourth Normal Form(4NF)
- ☐ **(B)** Third Normal Form(3NF)
- ☐ **(C)** Second Normal Form (2NF)
- ☐ **(D)** First Normal Form (1NF)

3. Which one of the following TCP/IP network configuration and troubleshooting command that used to ensure that a computer can communicate to a specified device over the network?

☐ **(A)** ping

☐ **(B)** ifconfig

☐ **(C)** ipconfig

☐ **(D)** Traceroute

4. Which of the following is the correct sequence of HTML tags to design a webpage?

☐ **(A)** HTML, Head, Title, Body

☐ **(B)** HTML, Head, Body, Title

☐ **(C)** HTML, Body, Title, Head

☐ **(D)** Head, Title, HTML, body

5. Which of the following is an application of stack?

- ☐ (A) Finding factorial
- ☐ (B) Infix to postfix conversion
- ☐ (C) Tower of Hanoi
- ☐ (D) All of the above

6. Which of the following data structure is linear type?

☐ **(A)** Array

☐ **(B)** Tree

☐ **(C)** Graphs

☐ **(D)** Hierarchy

7. In object-oriented programming, inheritance is one of the pillar for object oriented programming paradigm. So, which keyword is used to achieve the concept of "is a" on interface class?

☐ (A) Interfaces

☐ (B) Implements

☐ (C) Inherits

☐ (D) Extends

8. Which of the following is more advantage of POST method than GET?

- ☐ (A) Can be used to send ASCII as well as binary data.
- ☐ (B) Does not have any restriction on data size to be sent
- ☐ (C) All
- ☐ (D) Since data is sent through HTTP header, security depends on HTTP protocol

9. Which of the following is odd about constructor and method?

☐ **(A)** Constructor is a special type of method in the class

☐ **(B)** In OOP using java, Method over loading is possible, but constructor overloading is not

☐ **(C)** Like method, constructor have access or visibility modifier

☐ **(D)** Constructor have no return type, where as a method have

10. Which one of the following is a Linux system scripting syntax that used to create directory inside other specific directory?

☐ **(A)** `$mkdir dir_name`

☐ **(B)** `$mkdir{ dir_name1,
dir_name2, dir_name3}`

☐ **(C)** `$mkdir -p
dir_name1/dir_name2`

☐ **(D)** `$mkdir dir_name1/dir_name2`

11. Which part of memory can a computers' microprocessor access more quickly?

- ☐ **(A)** Virtual memory
- ☐ **(B)** Cache memory
- ☐ **(C)** Physical Memory
- ☐ **(D)** None

12. Which of these is not applicable for IP protocol?

- ☐ **(A)** Offer reliable service
- ☐ **(B)** Connectionless
- ☐ **(C)** Does not offer error reporting
- ☐ **(D)** Offer unreliable service

13. A bottom up parser generates _____?

- ☐ (A) Leftmost derivation in reverse
- ☐ (B) Right most derivation
- ☐ (C) Rightmost derivation in reverse
- ☐ (D) Leftmost derivation

14. In OOP approach to solve a problem, all computations are carried out using carried out on_____?

☐ **(A)** Attribute

☐ **(B)** Object

☐ **(C)** Class

☐ **(D)** Method

15. Which of the following is basic principle of computer programming?

- ☐ **(A)** Minimize coupling
- ☐ **(B)** Document your code
- ☐ **(C)** All of the above
- ☐ **(D)** Keep it simple

16. In a relational database the rule which can be described as “A Gender field can hold only the values M or F”. This is an example of___?

- ☐ **(A) Entity integrity constraint**
- ☐ **(B) Not Null integrity constraint**
- ☐ **(C) Referential integrity constraint**
- ☐ **(D) Domain integrity constraint**

17. Consider the following requirement for software: "The operator must be able to add new games without modifications to the existing system." Which one of the following types of requirements is this?

- ☐ **(A)** Reliability Requirement
- ☐ **(B)** Performance Requirement
- ☐ **(C)** Usability Requirement
- ☐ **(D)** Supportability Requirement

18. _____ Algorithm operates using a single current node and generally move only to neighbors of that node rather than systematically exploring paths from an initial state to find solution for search problem?

- ☐ (A) MinMax
- ☐ (B) None
- ☐ (C) Recursive best first search
- ☐ (D) Simulated annealing

19. The time complexity of quick sort is_____?

☐ **(A)** $O(n \log n)$

☐ **(B)** $O(\log n)$

☐ **(C)** $O(n^2)$

☐ **(D)** $O(n)$

20. Syntax-directed translation rules use_____?

- ☐ (A) None of the mentioned
- ☐ (B) Constants & attributes associated with the non-terminals in their definitions.
- ☐ (C) Lexical values of nodes
- ☐ (D) A & B

21. If you wanted to have 12 subnets with a Class C network ID, which subnet mask would you use?

☐ (A) 255.255.255.248

☐ (B) 255.255.255.252

☐ (C) 255.255.255.255

☐ (D) 255.255.255.240

22. Which of the following statement is true about forward chaining algorithm?

- ☐ **(A)** It is known as a top-down approach
- ☐ **(B)** It is a form of reasoning, which starts with the goal and works forward, chaining through rules to find known facts that support the goal.
- ☐ **(C)** In forward chaining, the goal is broken into sub-goal or sub-goals to prove the facts true
- ☐ **(D)** It is data driven algorithm.

23. _____ attempts not only to find a solution, but to find the ideal solution to any given problem.

- ☐ **(A)** Dynamic programming
- ☐ **(B)** Randomized algorithms
- ☐ **(C)** Greedy algorithms
- ☐ **(D)** Branch and bound algorithms

24. The logical or mathematical model of a particular organization of data is called a_____?

- ☐ **(A)** Data arrangement
- ☐ **(B)** Data formation
- ☐ **(C)** Data configuration
- ☐ **(D)** Data structure

25. Which of the following is not a notion of Context free grammars?

- ☐ (A) Derivations
- ☐ (B) Sentential forms
- ☐ (C) Recursive Inference
- ☐ (D) All of the mentioned

26. What is the purposes of using Node.js?

- ☐ (A) Distributed systems
- ☐ (B) All
- ☐ (C) Real-time web applications
- ☐ (D) General purpose applications

27. When illegally authorized finance director modify his own salary in a payroll database, thus which computer security objective is violated?

☐ (A) Confidentiality

☐ (B) Modification

☐ (C) Integrity

☐ (D) Availability

28. Which one is False about PHP?

- ☐ **(A)** PHP can't be directly embedded into HTML
- ☐ **(B)** PHP is an object-oriented language
- ☐ **(C)** PHP is a server-side scripting language and manage the dynamic content of the website
- ☐ **(D)** None

29. Polymorphism is one of the object oriented principle that an object is to have many form. This is achieved with statically or dynamically. The implementation of static polymorphism is using?

- ☐ **(A)** A & C
- ☐ **(B)** Method calling
- ☐ **(C)** Method overriding
- ☐ **(D)** Method overloading

30. Which malware spread through user interaction via downloading and running an executable file from the Internet?

- ☐ **(A) Viruses**
- ☐ **(B) Trojans**
- ☐ **(C) Spyware**
- ☐ **(D) Worms**

31. The subsystem in a good software design should have which of the following characteristics?

- ☐ **(A)** Low cohesion, low coupling
- ☐ **(B)** High cohesion, low coupling
- ☐ **(C)** High cohesion, high coupling
- ☐ **(D)** Low cohesion, high coupling

32. Let you have an IP address of 172.16.13.5 with a 255.255.255.128 subnet mask. What is your class of address, subnet address, and broadcast address?

☒ **(A)** Class B, Subnet 172.16.13.0,
Broadcast address 172.16.13.127

☐ **(B)** Class A, Subnet 172.16.13.0,
Broadcast address 172.16.13.127

☐ **(C)** Class B, Subnet 172.16.0.0,
Broadcast address 172.16.255.255

☐ **(D)** Class B, Subnet 172.16.13.0,
Broadcast address 172.16.13.255

33. In a priority queue, insertion and deletion takes place at_____?

- ☐ **(A)** any position
- ☐ **(B)** only at front end
- ☐ **(C)** front, rear end
- ☐ **(D)** only at rear end

34. How is an input–output instruction different from a register-reference instruction?

- ☐ **(A)** An input–output instruction does not need a reference to memory, whereas a register-reference instruction needs a reference to memory.
- ☐ **(B)** An input–output instruction is recognized by the operation code 111 with a 1 in the leftmost bit of the instruction, whereas a register-reference instruction is recognized by the operation code 111 with a 0 in the leftmost?

☐ **(C)** A and C

☐ **(D)** An input–output instruction does not need operand from memory, whereas a register-reference instruction needs operand from memory

35. ____describes inheritance relationship in the world of objects?

- ☐ **(A) Dependency**
- ☐ **(B) Generalization**
- ☐ **(C) Association**
- ☐ **(D) Realization**

36. Which kind of proof is used to prove the regularity of a language?

☐ **(A)** None of the mentioned

☐ **(B)** Proof by contradiction

☐ **(C)** Direct proof

☐ **(D)** Proof by induction

37. Which of the following is false about package?

- ☐ **(A)** It is common that to write package command as the first statement in a Java source
- ☐ **(B)** A program can be created and compiled without package
- ☐ **(C)** None
- ☐ **(D)** If the source code has no specified package name, the class names are put into the default

38. Calculate the First(S) and Follow(A) functions for the given grammar- $S \rightarrow AaAb / BbBa$ $A \rightarrow \epsilon$ $B \rightarrow \epsilon$

☐ (A) $\text{First}(S) = \{ a, b \}$ and $\text{Follow}(A) = \{ a, b \}$

☐ (B) $\text{First}(S) = \{ a, b \}$ and $\text{Follow}(A) = \{ a, \epsilon \}$

☐ (C) $\text{First}(S) = \{ a, \epsilon \}$ and $\text{Follow}(A) = \{ a, b \}$

☐ (D) $\text{First}(S) = \{ \epsilon, \epsilon \}$ and $\text{Follow}(A) = \{ a, b \}$

39. Suppose an employee of CBE who are familiar with only database interface, and operate a banking database system to pay and transfer money for their customers. Those employees are category of_____?

- ☐ **(A)** Advanced users'
- ☐ **(B)** Sophisticated users
- ☐ **(C)** Casual users
- ☐ **(D)** Naive users

40. Which of the following correctly declares an array in C++?

☐ **(A)** `int array[10];`

☐ **(B)** `int array;`

☐ **(C)** `array{10};`

☐ **(D)** `array array[10];`

41. Which statement is describing fragmentation in operating system?

- ☐ **(A)** All
- ☐ **(B)** Occurs when the memory is distributed into fixed-sized blocks.
- ☐ **(C)** It happens when processes cannot be allocated to memory blocks and considering their small size and memory blocks remain unused

☐ (D) Fragmentation occurs in a dynamic memory allocation system when many of the free blocks are too small to satisfy any request

42. Which handles dynamic content, database as well as session tracking for the website?

☐ **(A)** JavaScript

☐ **(B)** PHP

☐ **(C)** CSS

☐ **(D)** HTML

43. If the input of the prims algorithm is a weighted connected graph $G = (V, E)$ then its output is given by?

- ☐ (A) Set of edges comprising a binary tree
- ☐ (B) Set of edges comprising a connected graph
- ☐ (C) Set of vertices comprising a MST
- ☐ (D) Set of edges comprising a MST

44. From different exception handling mechanisms which one uses two block code writing format to hand exceptions under the method?

- ☐ **(A)** All
- ☐ **(B)** Throw
- ☐ **(C)** Try and catch
- ☐ **(D)** ML_CATCH()

45. The size of virtual memory is based on which of the following?

☐ **(A)** Address bus

☐ **(B)** CPU

☐ **(C)** Data bus

☐ **(D)** RAM

46. The grammar $S \rightarrow aSa \mid bS \mid c$ is_____?

- ☐ (A) LL(1) but not LR(1)
- ☐ (B) LR(1) but not LR(1)
- ☐ (C) None of the mentioned
- ☐ (D) Both LL(1) but not LR(1) & LR(1) but not LR(1)

47. Suppose you have one best a classmate, and you crack his user name and user password of OBU RIS via social engineering mechanism to his account through your mobile phone thus which fundamental computer security objective is compromised?

☐ **(A)** Confidentiality and Integrity

☐ **(B)** Confidentiality

☐ **(C)** Availability

☐ **(D)** Integrity

48. The most of government organizations that currently using Microsoft office Excel and MS-word applications to create, access and manage their organizational records. This record or data management approach is an example of ____?

- ☐ **(A) Database approach**
- ☐ **(B) Manual approach**
- ☐ **(C) Database Management system approach**

☐ **(D)** File based approach

49. Which of the following error can a compiler check?

- ☐ **(A)** Logical Error
- ☐ **(B)** Both Logical and Syntax Error
- ☐ **(C)** Compiler cannot check errors
- ☐ **(D)** Syntax Error

50. The best examples of recursive relationships in database management representation is

☐ **(A)** An employee who manages other employees

☐ **(B)** A course taken by students per semester

☐ **(C)** An employee who manages a department

☐ **(D)** A student assigned for departments

51. Purchasing insurance cover can be considered to be an example of which one of the following risk handling strategies____?

☐ (A) Acceptance

☐ (B) Avoidance

☐ (C) Transfer

☐ (D) Mitigation

52. In Linux system, which user that automatically created during the installation and has highest privileges to all services?

- ☐ **(A)** Normal user account
- ☐ **(B)** Service user account
- ☐ **(C)** Regular or normal user account
- ☐ **(D)** Super user account

53. _____Paradigm expresses computation by fully-specified and fully-controlled manipulation of named data in a stepwise fashion?

- ☐ **(A)** Applicative programming
- ☐ **(B)** Functional programming
- ☐ **(C)** Object-oriented programming
- ☐ **(D)** Procedural programming

54. Assume we use the Caesar cipher encryption mechanism to encrypt each alphabet letter, by encryption key of 17, thus, identifies M cipher text?

☐ **(A) W**

☐ **(B) D**

☐ **(C) C**

☐ **(D) F**

55. The effect of a successfully completed (committed) transaction are permanently recorded in the database and must not be lost because of subsequent failure indicates?

- ☐ **(A)** Atomicity
- ☐ **(B)** Consistency
- ☐ **(C)** Isolation
- ☐ **(D)** Durability

56. Computer architecture refers to_____?

- ☐ (A) The way in which computer components are interrelated
- ☐ (B) The operation of each individual component as part of the structure
- ☐ (C) The operational units of computer system
- ☐ (D) Attributes that have a direct impact on the logical execution of a

57. Which of the following OSI reference model layer is responsible for Logical Addressing and Routing?

- ☐ **(A) Network Layer**
- ☐ **(B) Data Link Layer**
- ☐ **(C) Transport Layer**
- ☐ **(D) Physical Layer**

58. A DPDA is a PDA in which_____?

- ☐ **(A)** More than one state can have two or more outgoing transitions
- ☐ **(B)** At least one state has more than one transition
- ☐ **(C)** None of the mentioned
- ☐ **(D)** No state p has two outgoing transitions

59. Which one of the following a security policies that safeguard traffic sensitive data that flows between the end-user from being exploited by cybercriminal?

- ☐ **(A) Logging**
- ☐ **(B) Encryption**
- ☐ **(C) Authentication**
- ☐ **(D) Authorization**

60. In which phases of a unified process, the project scope and boundary conditions are established?

- ☐ **(A)** Elaboration
- ☐ **(B)** Inception
- ☐ **(C)** Construction
- ☐ **(D)** Transition

61. The training is provided to the machine with the set of data that has not been labeled, and the _____ algorithm restructures the input data into new features or a group of objects with similar pattern?

- ☐ (A) B and C
- ☐ (B) Supervised learning
- ☐ (C) Reinforcement learning
- ☐ (D) Unsupervised learning

62. A hypothetical computer stores real numbers in floating point format in 16-bit words. The first bit is used for the sign of the number, the next 7 bits for the magnitude of the exponent, and the next 8 bits for the magnitude of the mantissa. The number 34.25 represented in base-2 is?

☐ **(A)** 0 1100110 10001001

☐ **(B)** 0 1010110 10001001

☐ **(C)** 0 1000110 10001001

☐ **(D)** 0 1010110 10001000

63. Which one of the following is a roles of system administrator?

- ☐ **(A)** Implement the backup and recovery security policy
- ☐ **(B)** User Administration and monitor system performance
- ☐ **(C)** Ensuring the systems are running efficiently and effectively and provide services to whole community rationally
- ☐ **(D)** All

64. The feature by which one object can interact with another object in object oriented database

☐ **(A) Message Passing**

☐ **(B) Data Transfer**

☒ **(C) Message reading**

☐ **(D) Data Binding**

65. Which HTML tag target attribute is opens the document in the same window or tab as it is clicked?

☐ **(A) _top**

☐ **(B) _self**

☐ **(C) _blank**

☐ **(D) _parent**

66. Which of the following database controls would ensure that the integrity of transactions is maintained in an online transaction processing system's database?

☐ **(A)** Commit and rollback controls

☐ **(B)** Authentication controls

☐ **(C)** Read/write access log controls

☐ **(D)** Data normalization controls

67. A digital computer has a common bus system for 16 registers of 32 bits each. The bus is constructed with multiplexers. How many selection inputs are there in each multiplexer?

☐ **(A) 6**

☐ **(B) 3**

☐ **(C) 4**

☐ **(D) 5**

68. Which of the following is not an example of finite state machine system?

☐ **(A)** Control Mechanism of an elevator

☐ **(B)** Combinational Locks

☐ **(C)** Digital Watches

☐ **(D)** Traffic Lights

69. Assume if you are network administrator which operating systems more suitable for purpose of management and provide services to the customer?

- ☐ **(A)** Network operating systems
- ☐ **(B)** Real Time operating system
- ☐ **(C)** Client side operating system
- ☐ **(D)** Desktop operating systems

70. Which of the following approach addresses the behavior of AI?

☐ **(A)** All

☐ **(B)** The "Laws of Thought" Approach

☐ **(C)** The Cognitive Modeling Approach

☐ **(D)** The Turing Test Approach

71. A list which displays the relationship of adjacency between elements is said to be?

☐ **(A)** linked list

☐ **(B)** Linear

☐ **(C)** nonlinear

☐ **(D)** trees

72. Which one of the following a countermeasures of computer security vulnerabilities?

- ☐ **(A)** Environmental monitoring and cryptographic techniques
- ☐ **(B)** All
- ☐ **(C)** Access control mechanisms and Security-awareness training
- ☐ **(D)** Web proxies & propose good security policies

73. _____ works by recursively breaking down a problem into two or more sub-problems of the same type, until these become simple enough to be solved directly?

- ☐ **(A)** Backtracking algorithms
- ☐ **(B)** Dynamic programming
- ☐ **(C)** Divide and conquer technique
- ☐ **(D)** Greedy algorithms

74. Which header file is used for reading and writing to a file?

- ☐ **(A) #include<fstream>**
- ☐ **(B) #include<file>**
- ☐ **(C) #include<fe>**
- ☐ **(D) #include<iostream>**

75. Which one of the following an operating systems peripheral regarding to security?

- ☐ **(A)** Linux operating system
- ☐ **(B)** UNIX Operating System
- ☐ **(C)** A and C
- ☐ **(D)** Window operating system

76. The activities involved in retrieving data from the database is ____?

- ☐ **(A)** Materialization
- ☐ **(B)** Pipelining
- ☐ **(C)** Query Optimization
- ☐ **(D)** Query Processing

77. The characteristics of knowledge based system that corresponds to the ability of manipulating the representational structures to produce new knowledge corresponding to existing structure is?

- ☐ **(A) Inferential Efficiency**
- ☐ **(B) Representational Accuracy**
- ☐ **(C) Acquisitional efficiency**
- ☐ **(D) Inferential Adequacy**

78. Among the major structural components of a computer, which component is responsible for controlling the operations of the computer?

- ☐ **(A) Central Processing Unit**
- ☐ **(B) Input/output**
- ☐ **(C) Main Memory**
- ☐ **(D) System interconnection**

79. _____ are used for serial transfer of data?

- ☐ **(A)** Register transfer micro-operations
- ☐ **(B)** Logic micro-operations
- ☐ **(C)** Arithmetic micro-operations
- ☐ **(D)** Shift micro-operations

80. The process of restoring the database to a state that is known to be correct after some failure has rendered the current state incorrect, or at least suspect is known as ____?

- ☐ **(A) Transaction**
- ☐ **(B) Concurrency control**
- ☐ **(C) Database backup**
- ☐ **(D) Database Recovery**

81. Which of the functions can a Turing machine not perform?

- ☐ **(A)** Copying a string
- ☐ **(B)** Accepting a pal
- ☐ **(C)** Deleting a symbol
- ☐ **(D)** Inserting a symbol

82. Which traversal technique processes the root node first?

- ☐ **(A)** In-order traversal
- ☐ **(B)** Preorder traversal
- ☐ **(C)** Post-order traversal
- ☐ **(D)** Level order traversal

83. The grammar $A \rightarrow AA \mid (A) \mid \epsilon$ is not suitable for predictive-parsing because the grammar is _____?

- ☐ (A) Ambiguous
- ☐ (B) An operator-grammar
- ☐ (C) Right-recursive
- ☐ (D) Left-recursive

84. What will be the output of the following C++ program? #include <iostream> using namespace std; int main() { int n = 5; void *p = &n; int *pi = static_cast<int*>(p); cout<<*pi << endl; return 0; }

☐ (A) Compile time error

☐ (B) 5

☐ (C) runtime error

☐ (D) 6

85. Why are external style sheets preferred?

- ☐ **(A)** Make the selectors and grouping to apply styles easily
- ☐ **(B)** All
- ☐ **(C)** Classes can be reused any number of times
- ☐ **(D)** It allows for the style control of multiple documents through a single file

86. Which of the following tag is used to add rows in the table?

☐ **(A)** <td> and </td>

☐ **(B)** <th> and </th>

☐ **(C)** None

☐ **(D)** <tr> and </tr>

87. Which Objects are in charge of realizing use cases?

☐ **(A)** Boundary object

☐ **(B)** A & B

☐ **(C)** Control objects

☐ **(D)** Entity objects

88. Which of the following is a condition that causes deadlock?

- ☐ **(A)** Mutual exclusion
- ☐ **(B)** Circular wait and No preemption
- ☐ **(C)** All
- ☐ **(D)** Hold and wait

89. Suppose $x=7$, $y=8$, $z=6$,
 $k=x+z*y/z\%x-z$ then what will be the
value of k after expression is evaluated?

☐ (A) 9

☐ (B) 2

☐ (C) 7

☐ (D) 8

90. Which is not true about Operating system ?

- ☐ **(A)** Application software that performs all basic tasks of a computer system.
- ☐ **(B)** Collection of programs that manages hardware resources
- ☐ **(C)** A program that acts as an intermediary between the user and the computer hardware
- ☐ **(D)** System service provider to the

91. Turing machine operates over____?

- ☐ **(A)** finite memory tape
- ☐ **(B)** None of the mentioned
- ☐ **(C)** infinite memory tape
- ☐ **(D)** depends on the algorithm

92. Which algorithm uses the backtracking method of recurring?

- ☐ **(A)** Greedy algorithms
- ☐ **(B)** Dynamic programming
- ☐ **(C)** Divide and conquer algorithms
- ☐ **(D)** Randomized algorithms

93. Which of the following routers use packet filtering?

☐ **(A)** Application Layer firewall

☐ **(B)** Packet filtering firewall

☐ **(C)** IP enabled firewall

☐ **(D)** Router enhance firewall

94. Which is true about scheduling in operating systems?

- ☐ (A) FCFS is the job that has been waiting the longest is served next.
- ☐ (B) The Shortest job First scheduling algorithm is a no preemptive scheduling algorithm that chooses the job that will execute the shortest amount of time.
- ☐ (C) All

○ **(D)** Round Robin scheduling is where each process gets a small quantity of time to run and then it is preempted and the next process gets to run

95. A _____ is a technique for finding a particular value in a sorted list.

☐ **(A)** Selection sort algorithm

☐ **(B)** Quicksort algorithm

☐ **(C)** Binary search algorithm

☐ **(D)** Merge sort algorithm

96. _____agents try to maximize their own expected happiness?

- ☐ (A) Goal-based
- ☐ (B) Simple reflex
- ☐ (C) Utility-based
- ☐ (D) Model-based reflex

97. Which of the following methods provides dedicated communications channel between two stations?

- ☐ **(A)** Switch network
- ☐ **(B)** Circuit switching
- ☐ **(C)** None of the above
- ☐ **(D)** Packet switching

98. A core command or language of DBMS used by end-users and programmers to store, retrieve, and access the data in the database is ____?

- ☐ (A) Data Definition Language (DDL)
- ☐ (B) Data Manipulation Language (DML)
- ☐ (C) Data Transaction Language (DTL)
- ☐ (D) Data Control Language (DCL)

99. Which one of the following is the best backup strategy for a large database with data supporting online sales?

- ☐ **(A)** Mirrored hard disks
- ☐ **(B)** Weekly full backup with daily incremental backup
- ☐ **(C)** Daily full backup
- ☐ **(D)** Clustered servers